



UNIVERSITI MALAYSIA TERENGGANU

GROUP PROJECT

CSF3233 - Cybersecurity

Topic: Email Security and Spam/Phishing Detection

GROUP 11

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2. PROJECT OVERVIEW

2.1 Introduction to the Project

Email has become one of the most widely used communication platforms in modern organizations, educational institutions, and businesses. However, its widespread use has also made it one of the primary targets for cyberattacks. Threats such as phishing, spear-phishing, email spoofing, and social engineering continue to evolve, causing significant financial losses, data breaches, and reputational damage worldwide. This project focuses on **Email Security and Spam/Phishing Detection**, with particular emphasis on understanding how attackers exploit both technical vulnerabilities and human behaviour to compromise systems and sensitive information. The selected sources highlight that cybersecurity is no longer solely a technical issue but also a human-centric challenge, as attackers increasingly rely on psychological manipulation to deceive victims.

To support this project, our group utilized NotebookLM as an AI-assisted academic tool to organize, analyze, and synthesize information from multiple credible sources. The platform allowed us to centralize our references, identify relationships between different studies, compare findings, and generate supporting materials that assisted our discussions and understanding of the topic. The analysis revealed several recurring cybersecurity concerns, including weaknesses in traditional email protocols, identity compromise, phishing attacks, and the increasing use of artificial intelligence by cybercriminals to create more sophisticated attacks. At the same time, various defensive mechanisms such as Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), Domain-based Message Authentication, Reporting and Conformance (DMARC), and security awareness training were identified as essential measures to strengthen email security.

Overall, this project demonstrates how AI tools such as NotebookLM can support academic research while reinforcing the importance of critical thinking, human verification, and responsible AI usage in cybersecurity education.

3. SOURCE SYNTHETICS

The project utilized six credible sources consisting of NIST technical publications, academic research papers, and an industry cybersecurity report. These sources were selected to provide a balanced understanding of email security from technical, human, and organizational perspectives. The NIST publications provided the foundational knowledge required to understand email security mechanisms and industry best practices. In particular, the guidelines explained how traditional email protocols contain inherent security weaknesses and emphasized

the importance of implementing authentication mechanisms such as Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), and Domain-based Message Authentication, Reporting and Conformance (DMARC). The NIST Phish Scale User Guide contributed a human-centred perspective by introducing a structured approach to measuring the difficulty of detecting phishing emails. This highlighted the importance of cybersecurity awareness training and demonstrated that human behaviour remains one of the most significant factors influencing cybersecurity incidents.

The Microsoft Digital Defense Report provided insights into current cybersecurity trends and emerging threats. The report revealed how cybercriminals and nation-state actors increasingly exploit identity-based attacks, artificial intelligence, and social engineering techniques to conduct large-scale cyber operations. In addition, the academic studies complemented the technical sources by exploring both behavioural and preventive approaches. The study by Butavicius and Parsons examined how attackers exploit psychological principles such as authority, scarcity, and social proof to deceive users through phishing campaigns. Meanwhile, Nmachi proposed a framework that focuses on strengthening email security by implementing additional authentication measures and advanced detection techniques. Collectively, these sources complement one another by covering different dimensions of cybersecurity. While some sources focus on technical infrastructures and security protocols, others emphasize human vulnerabilities, attacker behaviour, and emerging technologies. The combination of these perspectives allowed our group to develop a more comprehensive understanding of email security and phishing detection rather than relying on a single defensive approach.

Overall, the selected sources demonstrate that effective cybersecurity requires a multidisciplinary strategy that integrates technical safeguards, user awareness, organizational policies, and continuous adaptation to evolving threats.

4. EVIDENCE OF NOTEBOOKLM USAGE

NotebookLM was used as the primary AI-assisted academic support tool throughout this project. The platform allowed our group to centralize all research materials, analyze multiple sources simultaneously, and generate supporting outputs that improved our understanding of email security and phishing detection.

Initially, all selected references were uploaded into a shared NotebookLM notebook to create a centralized knowledge base. The uploaded materials included NIST publications, academic research papers, and industry cybersecurity reports. Organizing these sources within a single workspace allowed our group to easily access information and establish connections between different studies.

4.1 Notebook Setup

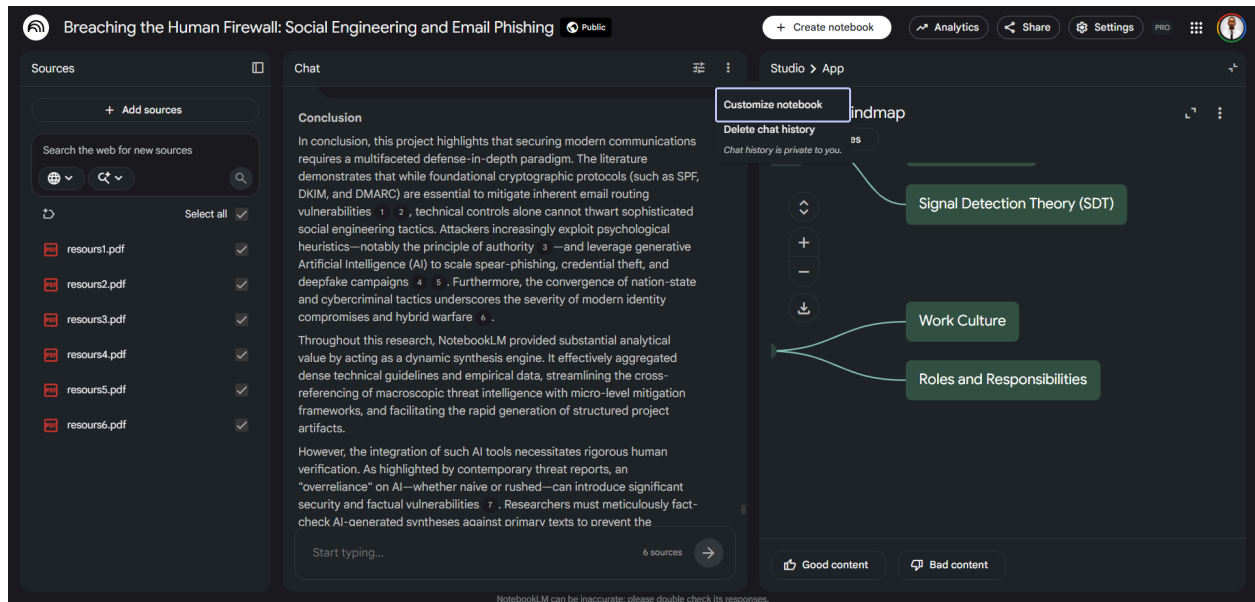


Figure 1: NotebookLM workspace showing the project notebook and uploaded sources.

NotebookLM was then used to summarize technical information and identify major cybersecurity concerns discussed across all sources. This helped our group quickly understand recurring themes, including phishing attacks, identity compromise, email spoofing, and social engineering techniques.

4.2 Prompts Used in NotebookLM

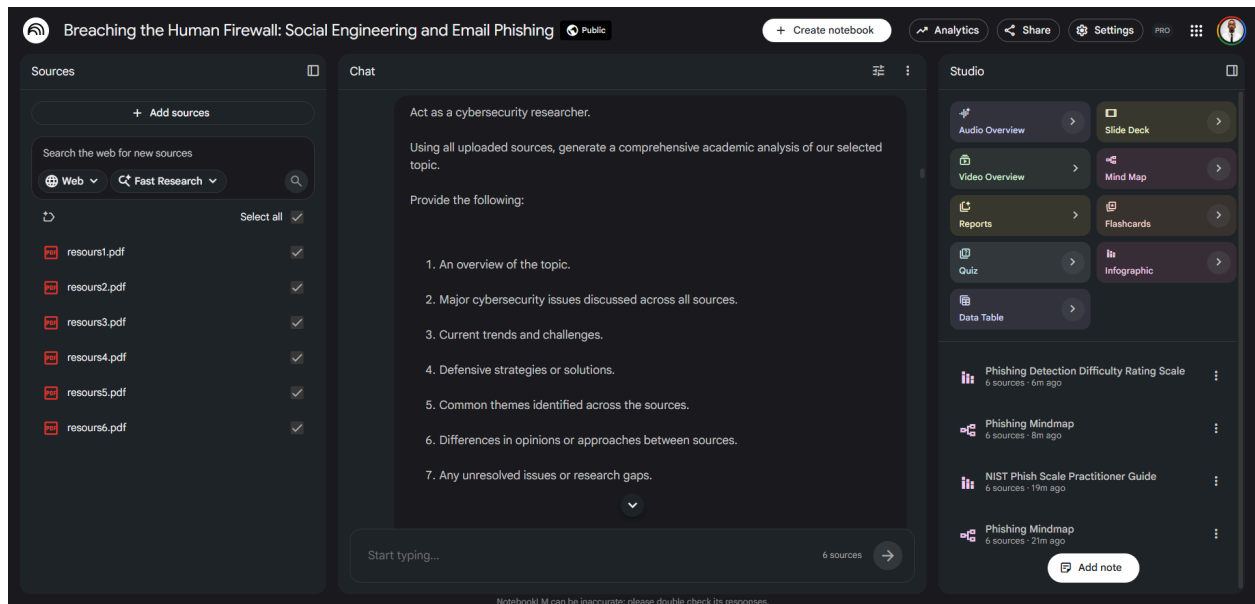


Figure 2: Examples of prompts used to interact with NotebookLM.

The platform was also used to compare findings from different sources. By analyzing multiple references simultaneously, NotebookLM helped identify similarities, differences, and relationships between technical security measures, human behaviour, and current cybersecurity trends.

4.3 Generated Responses

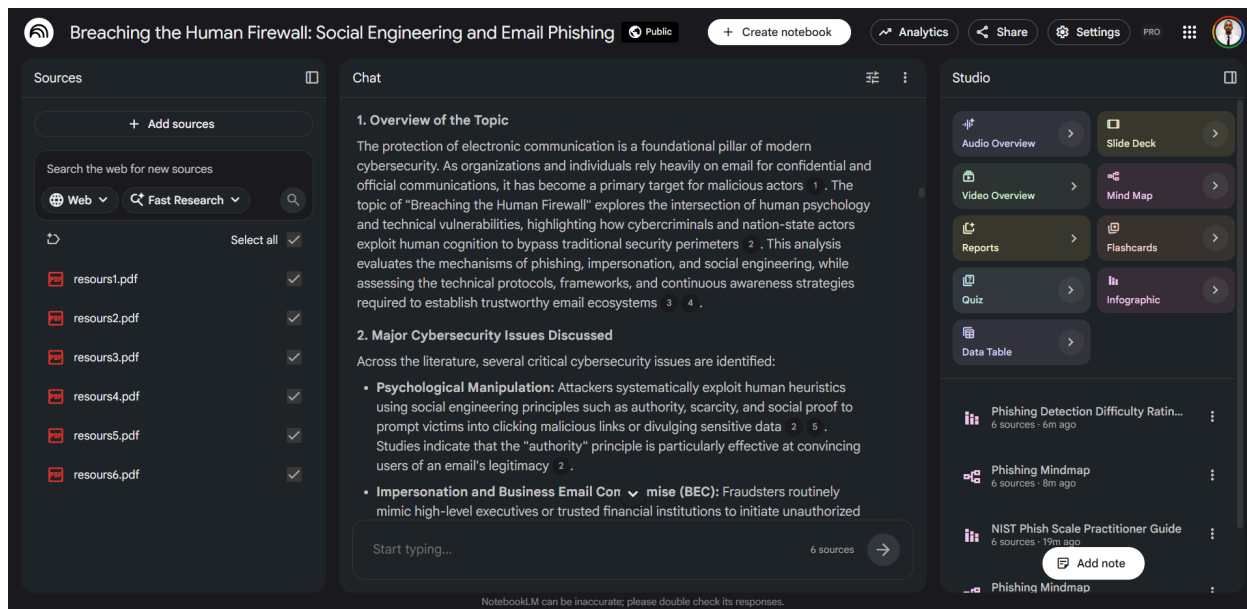


Figure 3: Example responses generated by NotebookLM.

In addition, NotebookLM assisted the group in generating supporting materials such as summaries, discussion points, and project artifacts. These outputs improved group discussions and provided a clearer understanding of the selected cybersecurity topic.

Overall, NotebookLM significantly improved the efficiency of the research process. However, all generated outputs were manually reviewed and verified against the original sources before being included in the final project to ensure accuracy and maintain academic integrity.

5. NotebookLM ARTIFACTS SUMMARY

As part of the project requirements, our group generated two NotebookLM artifacts to support the understanding and presentation of the selected topic, **Email Security and Spam/Phishing Detection**. These artifacts transformed information gathered from multiple sources into visual learning materials that simplified complex cybersecurity concepts and improved knowledge sharing among group members.

5.1 Artifact 1: Phishing Analysis and Detection Difficulty Mind Map

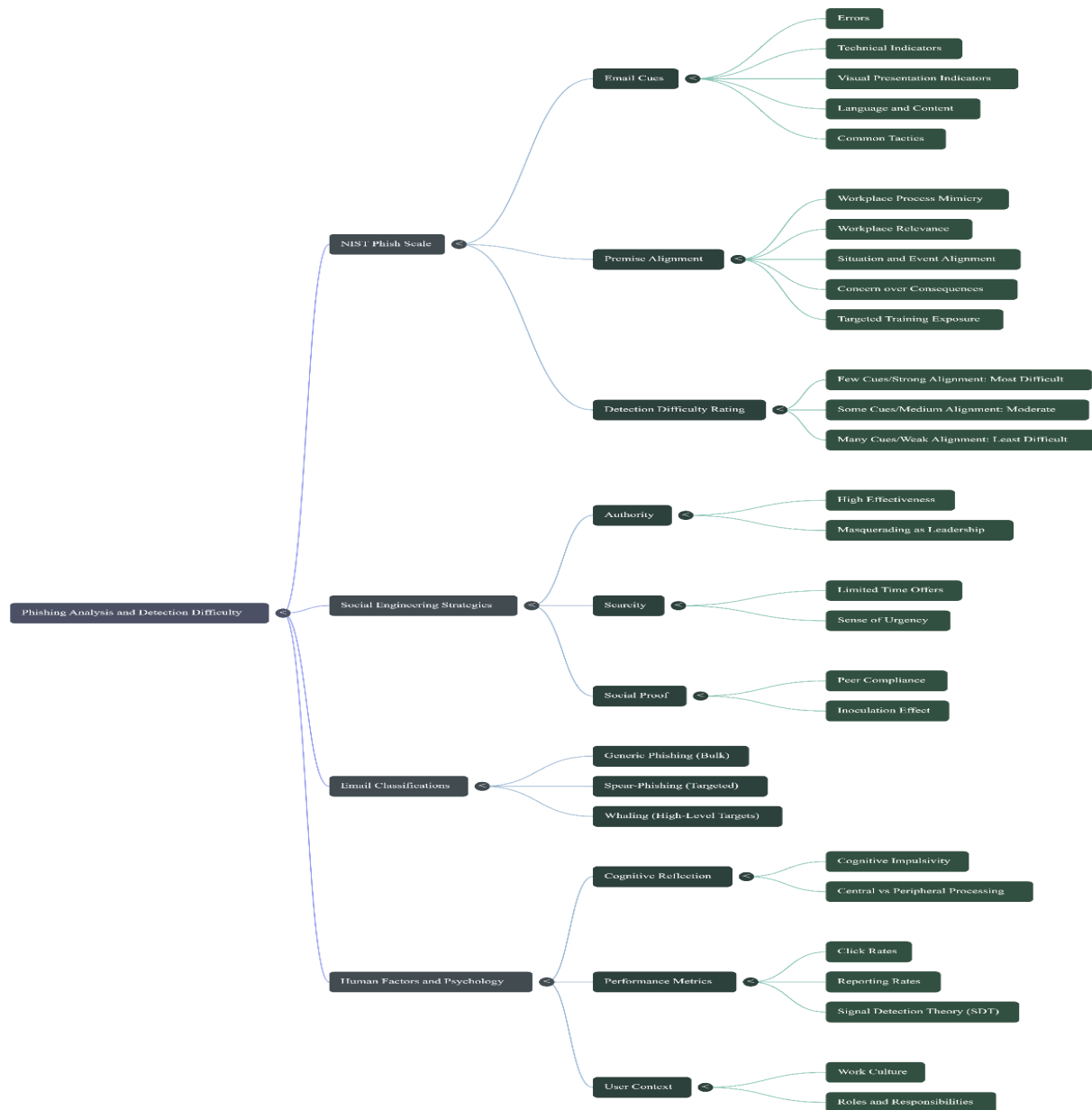


Figure 4: Mind map illustrating the relationships between phishing detection, social engineering strategies, email classifications, and human factors.

This mind map provides a structured overview of the major concepts associated with phishing attacks and detection difficulty. The artifact organizes the information into four major areas: the NIST Phish Scale, social engineering strategies, email classifications, and human factors that influence phishing susceptibility.

The visualization highlights how attackers exploit psychological principles such as authority, scarcity, and social proof to manipulate victims. It also categorizes different forms of phishing

attacks, including generic phishing, spear-phishing, and whaling attacks. In addition, the artifact demonstrates that human behaviour, workplace context, and cognitive decision-making significantly influence phishing detection outcomes.

The mind map helped our group quickly identify relationships between different concepts and served as a useful reference during project discussions and preparation.

5.2 Artifact 2: The NIST Phish Scale – A Practitioner's Guide to Detection Difficulty

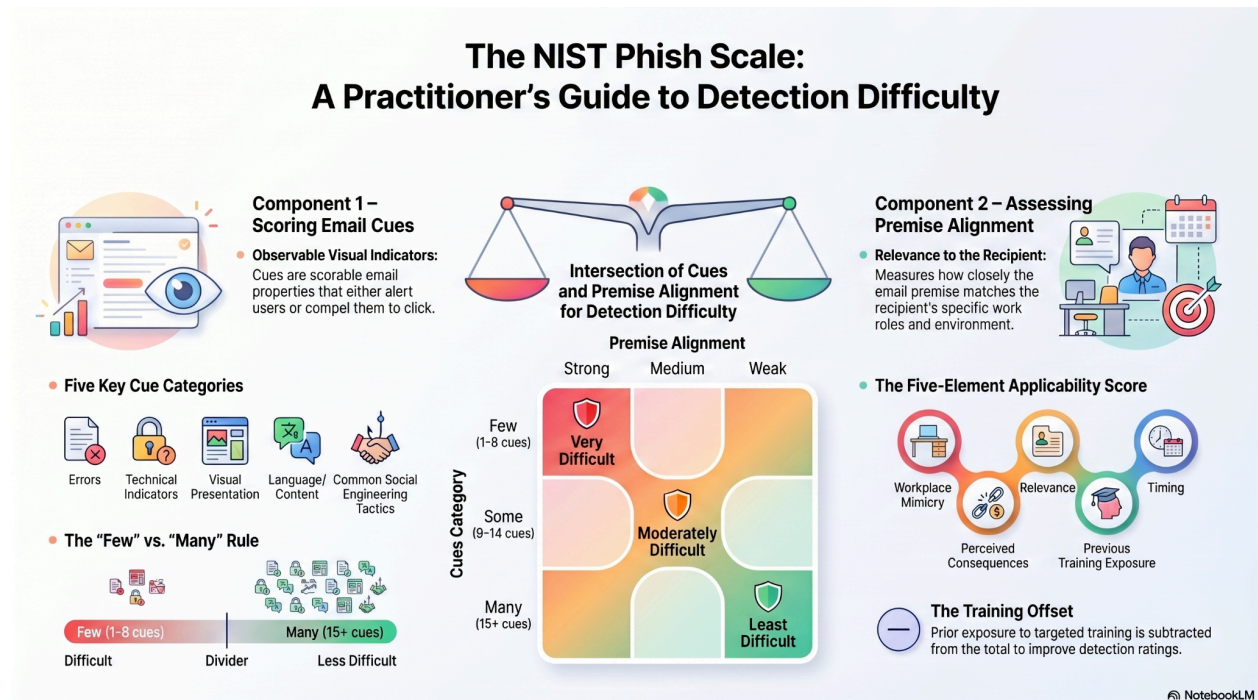


Figure 5: Infographic explaining the components of the NIST Phish Scale used to assess phishing detection difficulty.

This infographic provides a simplified explanation of the NIST Phish Scale, a framework used to assess how difficult it is for users to detect phishing emails. The artifact summarizes two key evaluation components: email cues and premise alignment.

The infographic explains how various indicators, such as visual presentation, technical indicators, language patterns, and social engineering tactics, contribute to phishing detection difficulty. It also demonstrates how workplace relevance, perceived consequences, and previous training exposure influence a user's ability to recognize phishing attempts.

By converting technical information into a visual format, this artifact made the NIST Phish Scale easier to understand and communicate. It also highlighted the importance of combining technical controls with continuous cybersecurity awareness training.

Overall, both artifacts improved the effectiveness of the project by transforming complex research findings into concise and accessible visual materials that supported group discussions, report preparation, and knowledge retention. The complete artifact files are included separately in the **04_Artifacts** folder.

6. REFLECTION SUMMARY

Throughout this project, NotebookLM played an important role in improving our research workflow and helping us manage multiple cybersecurity sources more efficiently. Instead of manually reviewing each document repeatedly, our group used the platform to organize references, summarize technical information, and identify connections between different studies related to email security and phishing detection.

One of the biggest advantages was the ability to quickly compare findings from various sources. This allowed us to identify common cybersecurity issues, emerging trends, and effective defensive strategies within a shorter period of time. NotebookLM also assisted in generating visual artifacts and supporting materials that improved our understanding of the topic and facilitated group discussions.

However, the project also highlighted several limitations of AI-assisted tools. Some generated outputs lacked context, oversimplified technical concepts, or required further clarification. As a result, manual review remained a necessary part of the research process. Our group continuously cross-checked important information against the original references before incorporating it into the final submission.

Overall, this project demonstrated that NotebookLM is most effective when used as a support tool rather than a replacement for human analysis and critical thinking. It improved productivity, simplified literature analysis, and enhanced collaboration within the group while reinforcing the importance of responsible AI usage in academic work.

A complete group reflection is provided separately in the **05_Reflection** folder.

7. MANUAL VERIFICATION SUMMARY

To ensure the reliability and academic quality of this project, all NotebookLM-generated outputs were manually reviewed and verified against the original uploaded sources before being included in the final submission.

The verification process focused on validating important statistical claims, technical explanations, phishing detection frameworks, and cybersecurity recommendations generated by the AI tool. Our group cross-checked this information using the original NIST publications, academic research papers, and industry reports to confirm their accuracy and relevance.

During the review process, several statements were refined to provide additional context, prevent overgeneralization, and improve technical accuracy. This was particularly important for statistical findings, experimental results, and explanations of security mechanisms, as AI-generated outputs may occasionally oversimplify complex cybersecurity concepts.

This verification process reinforced the importance of maintaining human oversight when using AI-assisted academic tools. Rather than treating AI-generated content as a final source of information, our group used NotebookLM as a support system while relying on manual validation to preserve academic integrity and ensure that the final work accurately reflected our understanding of the topic.

A detailed manual verification document containing the verification records, reviewed claims, and corrections made after validation is provided separately in the **06_Verification** folder.

8. Conclusion

This project explored the growing challenges associated with email security, phishing attacks, and social engineering in today's cybersecurity landscape. The analysis of multiple credible sources demonstrated that email remains one of the most targeted communication platforms due to both technical vulnerabilities and human factors that attackers can exploit. The findings showed that modern cyber threats are becoming increasingly sophisticated, particularly with the integration of artificial intelligence into phishing campaigns and identity-based attacks. As a result, organizations can no longer rely solely on technical security measures. Effective protection requires a combination of email authentication protocols, strong identity protection mechanisms, continuous security awareness training, and informed user behaviour.

NotebookLM significantly supported the overall research process by helping our group organize information, compare findings from different sources, and generate useful supporting materials. The platform improved efficiency and simplified the process of understanding complex cybersecurity concepts by transforming large amounts of information into structured and accessible outputs. At the same time, this project reinforced the importance of human oversight when using AI-assisted academic tools. Although NotebookLM provided valuable support, manual verification remained essential to ensure accuracy, maintain academic integrity, and prevent the misuse of AI-generated information.

Overall, this project demonstrated that responsible AI usage, critical thinking, and human judgement are equally as important as technical cybersecurity measures. When used appropriately, AI tools such as NotebookLM can enhance learning, improve collaboration, and support academic research while still requiring active human participation throughout the process.

References

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